

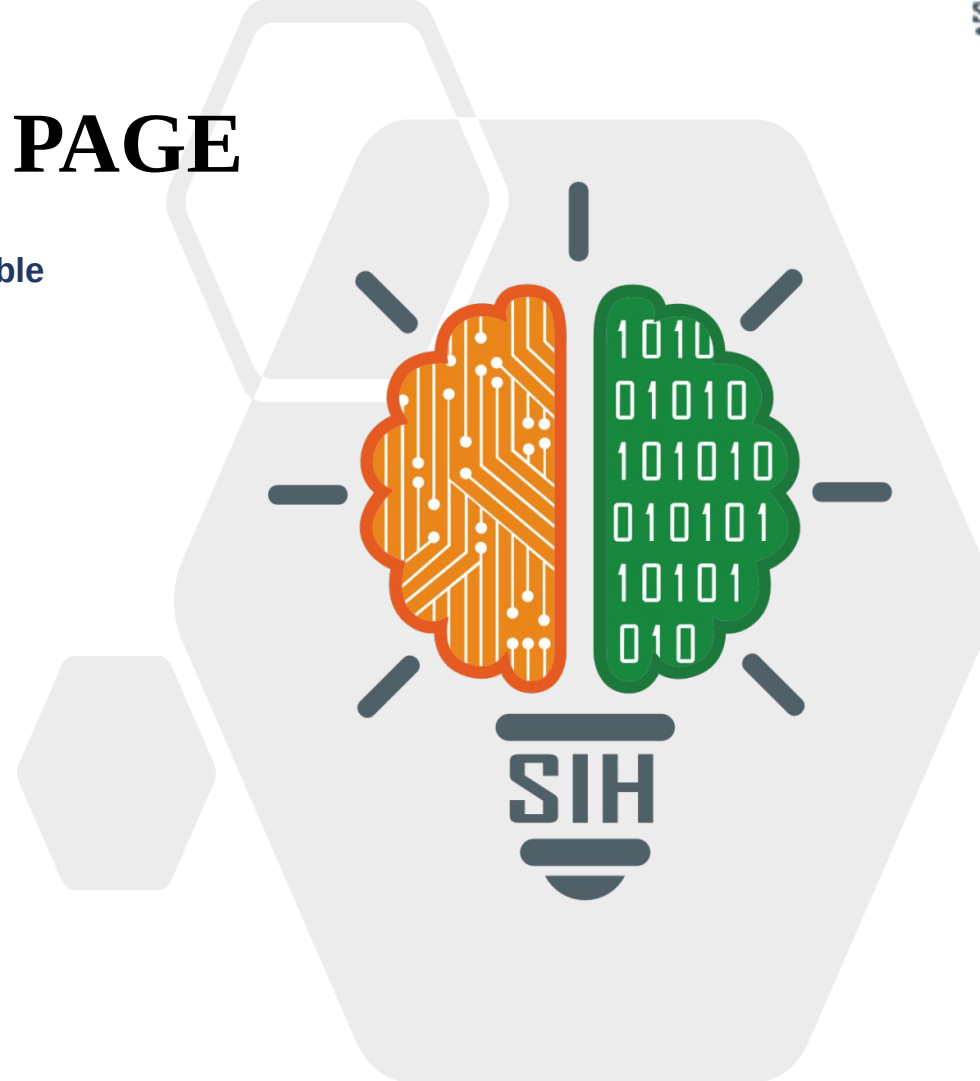
SMART INDIA HACKATHON 2025



TITLE PAGE

Idea: YatraDrishti – Hyper-Personalized AI Ecosystem for Sustainable Tourism & Smart Hospitality

- **Problem Statement ID** – _____
- **Problem Statement Title**- Student Innovation
- **Theme**- Travel & Tourism
- **PS Category**- Software
- **Team ID**- _____
- **Team Name (Registered on portal)** _____



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

❖ Detailed explanation of the proposed solution

YatraDrishti is a multi-sided AI platform that unites **tourists, hoteliers, transport providers and local artisans** in one intelligent ecosystem: a Sustainable Tourism Operating System that links demand for hotels with predictive crowd balancing for governments.



❖ How it addresses the problem

- **Overcrowded tier-1 hotspots** → predictive, time-shifted rerouting
- **15–30% OTA commissions** → zero-commission direct booking engine
- **Guesswork hotel pricing** → AI demand forecast & dynamic rates
- **Invisible artisans & homestays** → verified marketplace inside itineraries
- **Static, unsafe itineraries** → weather-aware plans + offline SOS

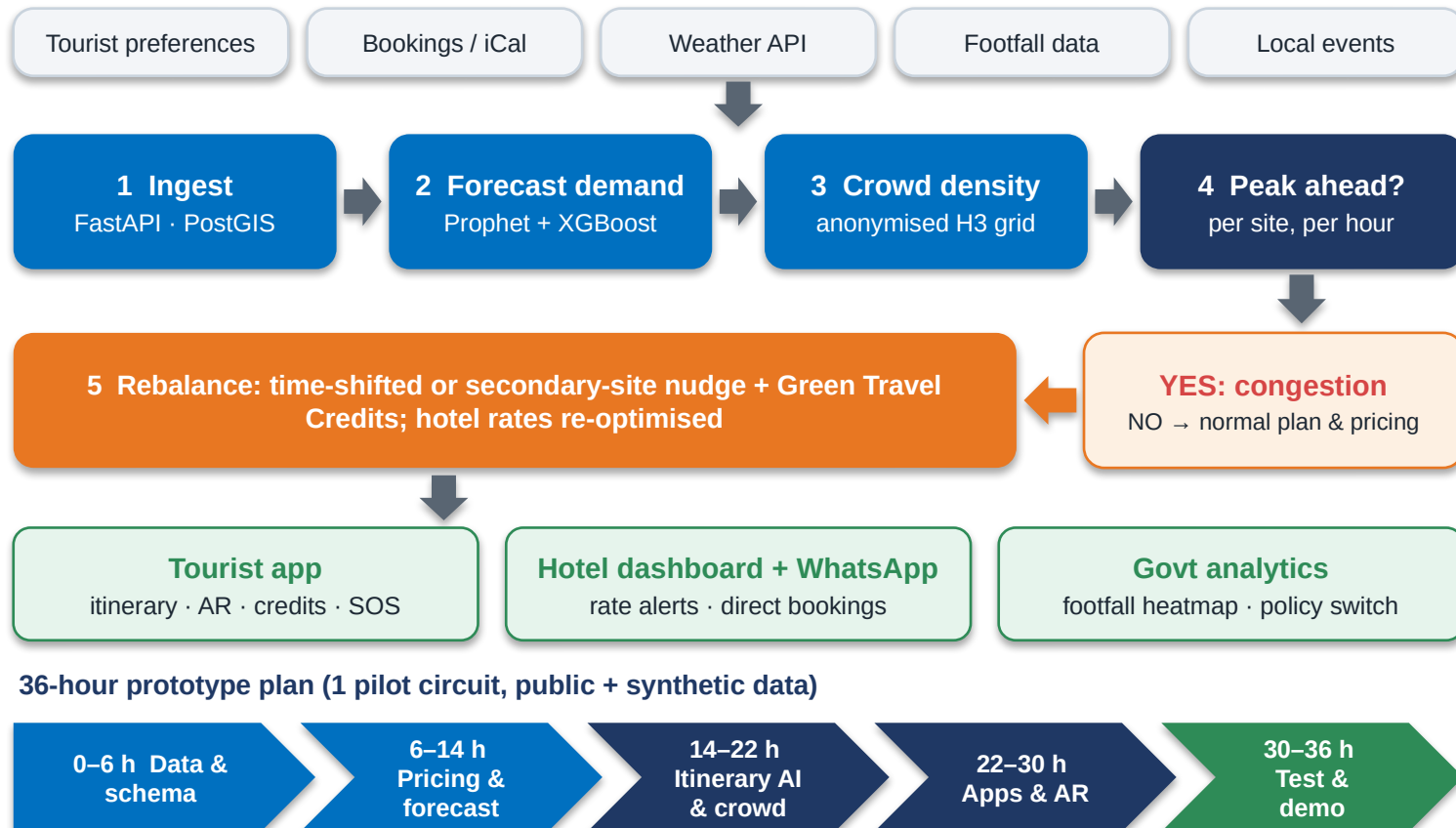
❖ Innovation and uniqueness of the solution

- **Tourism OS:** tourist actions update hotel & vendor analytics live
- **Green Travel Credits:** crowd pressure becomes revenue for nearby spots
- **Enterprise AI for small hotels:** 5-minute setup via WhatsApp or iCal
- **Offline Edge-AI:** safety guide works in no-network regions
- **Not an aggregator or GPT wrapper:** real pricing models + spatial analytics

Technologies to be used (e.g. programming languages, frameworks, hardware)

Layer	Technology
Frontend	React Native / Flutter · Next.js dashboard
Backend	FastAPI (Python) · Node.js (TypeScript) gateway
WhatsApp	Multi-tenant Baileys AI gateway (already built)
AI / ML	XGBoost + Prophet · PyTorch · LangChain + Llama-3 / Gemini
Data	PostgreSQL + PostGIS · Redis · H3 hexagon grid
AR / Edge	ARCore / ARKit · TensorFlow Lite (on-device)
APIs	OpenStreetMap · OpenWeatherMap · data.tourism.gov.in · Razorpay (test)
Cloud	Docker · Kubernetes · AWS / GCP
Hardware	None: runs on any smartphone

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Emergency policy switch: one admin flag freezes discounts and reroutes tourists during flood / weather alerts (can be shown live to judges).

Analysis of the feasibility of the idea



Technical ✓

- Multi-tenant WhatsApp AI gateway already working
- Per-tenant AI prompts defined



Data & APIs ✓

- data.tourism.gov.in, OSM, weather, Kaggle hotel data
- Backup: synthetic footfall + 50 hotels



Economic ✓

- Hotel SaaS ≈ ₹1,250–4,200 / month
- 3–5% fee on local experiences



Operational ✓

- 5-minute hotel setup via WhatsApp / iCal
- Fits existing hotel PMS

Phase 1: Pilot: Golden Triangle, Puducherry, Himachal

Phase 2: State Tourism Dev. Corporations

Phase 3: Hotel associations & homestay networks

National: Incredible India integration

Potential challenges and risks

Strategies for overcoming these challenges

WhatsApp session keys exposed in code

Revoke linked device, purge .baileys_auth from git, secrets in environment variables

Sessions lost on every restart

Database-backed auth store (PostgreSQL / Redis) with caching & safe key serialisation

WhatsApp rate limits / number ban

Outbound queue, rate limiter, randomised delays; official WhatsApp Business API in production

Cross-tenant data leakage

Per-tenant session factory + JWT tenant binding on every endpoint

No live telecom crowd data

Anonymised H3 density counts + Poisson-simulated footfall (DPDP Act 2023)

Hotels resist new software

Zero-friction iCal / webhook sync with existing PMS; rate tips on WhatsApp

❖ Potential impact on the target audience



Tourists

20–35%

savings on off-peak routes*

- Budget-aware custom trips
- Eco-travel rewards
- Offline SOS



Small hotels

+22–30%

low-season yield*

- Zero-commission bookings
- AI rate tips on WhatsApp
- Own their guest data



Artisans & locals

+35%

direct revenue*

- Verified digital badge
- Direct tourist reach
- No middlemen



Governments

up to 40%

less peak congestion*

- Predictive heatmaps
- Emergency policy switch
- Greener destinations

❖ Benefits of the solution (social, economic, environmental, etc.)



Social

- Preserves indigenous crafts & heritage
- Brings tier-2/3 & rural areas into tourism
- Multilingual access for all



Economic

- Spend shifts from OTAs to local economies
- Hotels keep 100% of direct-booking margin
- New income for guides & homestays



Environmental

- Less strain on fragile spots (Ladakh, Shimla, Munnar)
- Rewards off-peak & eco-friendly transit
- Supports SDG 8 & SDG 12



Safety & privacy

- Offline SOS in no-network areas
- DPDP Act 2023: no PII, anonymised H3 counts
- Weather-aware re-planning

*Target outcomes to be validated in the pilot; microservice-based dynamic pricing has shown +22% revenue in published research (Barua & Kaiser, 2024).

Details / Links of the reference and research work



Data, platforms & policy

- Ministry of Tourism data portal (tourist footfall) data.tourism.gov.in
- OpenStreetMap (heritage sites, transit nodes) openstreetmap.org
- OpenWeatherMap API (weather-aware plans) openweathermap.org
- Antonio et al. (2019) Hotel booking demand datasets, Data in Brief 22 [doi:10.1016/j.dib.2018.11.126](https://doi.org/10.1016/j.dib.2018.11.126)
- Baileys WhatsApp Web library (our gateway) github.com/WhiskeySockets/Baileys
- Digital Personal Data Protection Act 2023 meity.gov.in
- UN Sustainable Development Goals 8 & 12 sdgs.un.org/goals



Research papers

- Barua & Kaiser (2024) Microservices dynamic pricing in travel: +22% revenue, 17% faster pricing [arXiv:2411.01636](https://arxiv.org/abs/2411.01636)
- Gössling & Mei (2025) AI and sustainable tourism: risks & opportunities for the SDGs, Current Issues in Tourism [doi:10.1080/13683500.2025.2477142](https://doi.org/10.1080/13683500.2025.2477142)
- Sidiq et al. (2025) Role of AI in transforming smart tourism, JSEIT [doi:10.31764/jseit.v5i2.30705](https://doi.org/10.31764/jseit.v5i2.30705)
- Chen & Guestrin (2016) XGBoost: a scalable tree boosting system, KDD [doi:10.1145/2939672.2939785](https://doi.org/10.1145/2939672.2939785)
- Taylor & Letham (2018) Forecasting at scale (Prophet), The American Statistician [doi:10.1080/00031305.2017.1380080](https://doi.org/10.1080/00031305.2017.1380080)

Our prototype base: working multi-tenant WhatsApp AI gateway (Node.js + Baileys) with per-tenant AI prompts; codebase security review done and remediation plan defined (see Feasibility).

Competitors studied: MakeMyTrip, Booking.com, EaseMyTrip (OTAs, 15–30% commission, no crowd balancing) · Mindtrip.ai (AI planner, no hotel/PMS link) · Airbnb Experiences · state tourism apps (information only).